

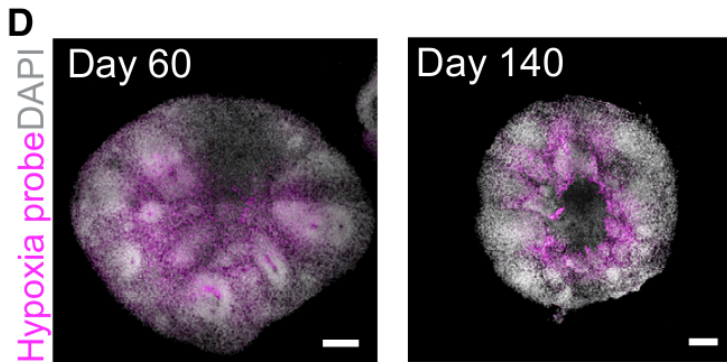
# Growing tissues and beyond

Jakub Koubele

Beyer lab retreat

9th September, 2026

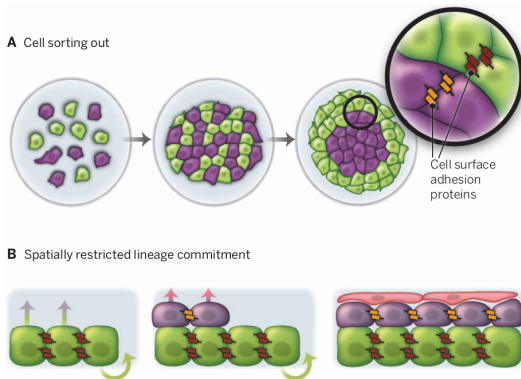
# Oxygen supply



Scale bar =  $200\ \mu\text{m}$ . Figure 1D from Qian et al. (2020). Note that day 140 is zoomed out, as the organoid grows over time.

- Oxygen diffuses only up to  $200\ \mu\text{m}$  in tissues.
- A vascular system to deliver oxygen and nutrients is therefore necessary for tissue thicker than  $\sim 400\ \mu\text{m}$ .

# Self-organization of cells



**Principles of self-organization.** Figure 2 from Lancaster and Knoblich (2014).

- Cells sort themselves by adhesion.
- However, vessels require endothelial cells (a separate lineage).
- Active blood flow is also needed (vessels regress otherwise).

# Vascularization of organoids

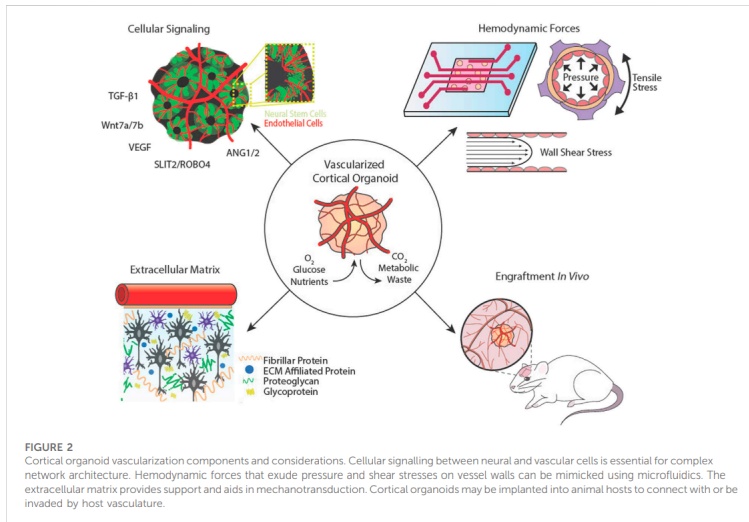
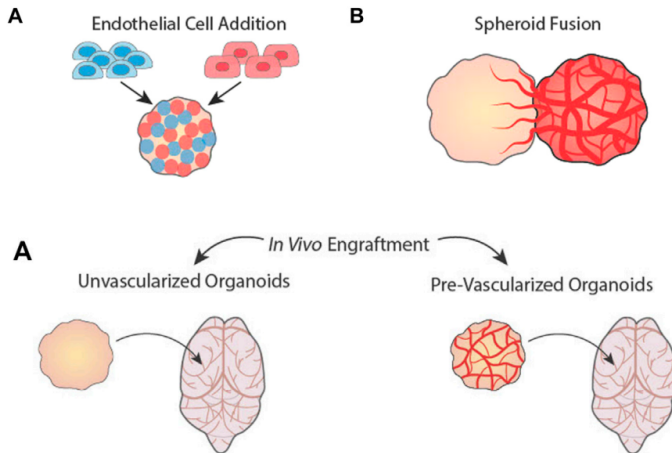


Figure 2 from LaMontagne et al. (2022).

# Vascularization of organoids (cont.)



Figures 3 and 5 from LaMontagne et al. (2022).

# Bioprinting

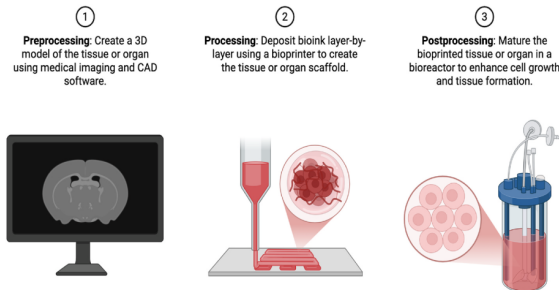


Fig. 1. Schematic representation of the 3D bioprinting workflow. Created with [Biorender.com](https://biorender.com).

Figure 1 from Idnani and Khan (2026).

- 3D printing approach. Bioink = differentiated cells + ECM components.
- Need lots of cells (costly and challenging to produce).
- Trade-offs between resolution, viscosity, speed and cell viability.

# Vascularization with bioprinting

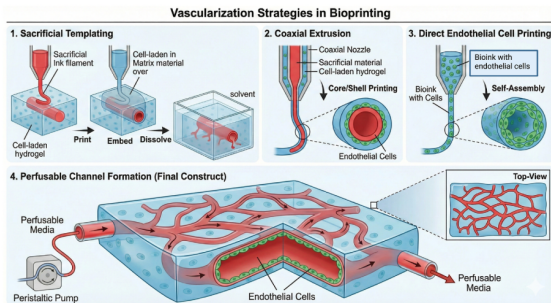
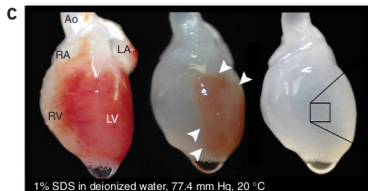


Fig. 5. Process-level vascularization strategies for perfusable bioprinted tissue. Created with [Biorender.com](https://biorender.com).

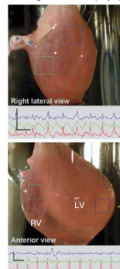
Figure 5 from Idnani and Khan (2026).

- Bioprinting approaches help with vascularization, but the resolution is still far away from real capillaries.

# Decellularized matrix



Beating recellularized (Day 4)

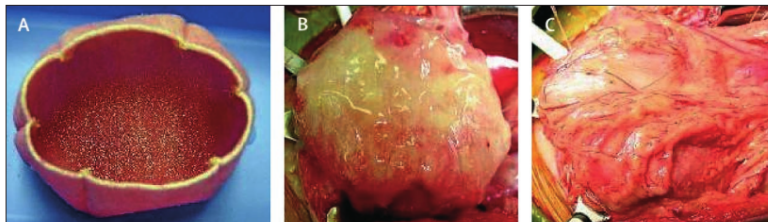


Figures 1c and 4d from Ott et al. (2008).

- Paper *Perfusion-decellularized matrix: using nature's platform to engineer a bioartificial heart*.
- Remove cells from a rat heart, keep the ECM. The vascular network is preserved.
- Repopulate with cells → it beats (but only  $\sim 2\%$  of normal heart function).



# Engineered bladder



**Figure 1: Construction of engineered bladder**

Scaffold seeded with cells (A) and engineered bladder anastomosed to native bladder with running 4-0 polyglycolic sutures (B). Implant covered with fibrin glue and omentum (C).

Figure 1 from Atala et al. (2006).

- 2006: seven patients, their own bladder cells expanded onto a collagen-PGA scaffold.
- The host grows the blood supply into it. Worked because a bladder is thin, hollow and metabolically cheap.

# Trachea scandal

## Clinal transplant of a tissue-engineered airway

Paolo Macchiarini, Philippe Jungbluth, Tetsuhiko Gu, M Adilade Aungmy, Louisa F Ross, Tostan A Cogner, Amanda Dodson, Joanne Montiel, Silvia Bellini, Pier Paolo Paragetti, Sally C Dickinson, Anthony P Hollander, Sara Morten, Maria Teresa Canani, Martin A Birchall

### Summary

**Background** The loss of a normal airway is devastating. Attempts to replace large airways have met with various problems. Prerequisites for a tissue-engineered replacement are a suitable matrix, cells, ideal mechanical properties, and the absence of antigenicity. We aimed to bioengineer tubular tracheal matrices, using a tissue-engineering protocol, and to assess the application of this technology in a patient with end-stage airway disease.

**Methods** We removed cells and MHC antigens from a human donor trachea, which was then seeded with cultured human epithelial cells and mesenchymal stem-cell-derived chondrocytes that had been cultured on collagen from a recipient (a 30-year-old woman with end-stage bronchomalacia). This graft was then used to replace the trachea of the left main bronchus.

**Findings** The graft immediately provided the recipient with a functional airway, improved quality of life, and had a normal appearance and mechanical properties at 4 months. The patient was free of donor antigens and was not on immunosuppressive drugs.

**Interpretation** The results show that we can produce a cellular, tissue-engineered airway with mechanical properties that allow normal functioning, and which is free from the risks of rejection. Our findings suggest that autologous cells combined with appropriate biomaterials might provide a useful treatment for patients with serious clinical disorders.

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### Introduction

Large airway defects present a major problem for clinicians because of the absence of any effective medical treatment. Tracheal resection and repair is the only curative treatment for benign or malignant processes, but covers the resected length is restricted to 30% of total length in children or 6 cm in adults, and myeloid tracheal defects will only be feasible if a safe, functional tracheal replacement can be developed. Tissue-engineered autologous or synthetic tracheal replacement has been developed.

Tissue engineering has provided functional human organ replacements elsewhere.<sup>1–5</sup> Previous preclinical airway replacements have been too lengthy and complex for routine clinical application,<sup>6</sup> or relied on

### Methods

**The recipient** A 30-year-old woman presented in 2004 with dysphonia and cough due to tuberculous infiltration of the cervical trachea and entire left main bronchus. A CT scan showed a circumferential, non-obstructive 3 cm airway stenosis starting 2 cm subglottically; and a hypoplastic left main bronchus with respiratory collapse. The mycobacterial infection was successfully treated over the next 6 months but severe dyspnoea persisted. Histology showed squamous metaplasia without residual infection, and we made a diagnosis of post-tuberculous chronic tracheitis and secondary severe bronchomalacia of the left main bronchus. Subsequently, the tracheal stenosis was successfully treated with a subglottic resection with primary end-to-end anastomosis, followed by placement

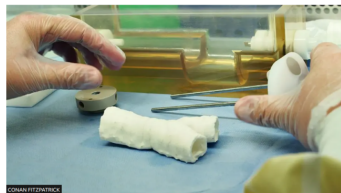


doi:10.1016/j.lan.2013.09.001

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## Paolo Macchiarini: A surgeon's downfall

10 September 2016

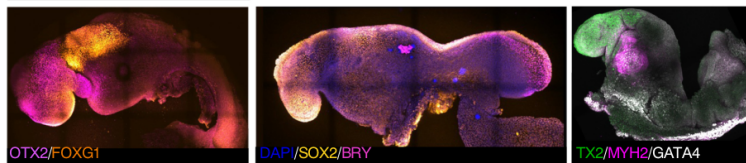


CONAN FITZPATRICK

A synthetic trachea before being placed in a bioreactor

- 2008–2013: tracheas (decellularized, later synthetic) seeded with the patients' own cells. Most recipients died.
- No blood supply — a trachea has no single artery, just many small vessels.
- Four *Lancet* papers retracted; Macchiarini convicted of gross assault in Sweden (2023).

# Synthetic embryos



Mouse embryo models can develop to the organogenesis stage. Figure 5C from Fu et al. (2025).

- Built from stem cells alone (no egg/sperm).
- Reach only E8.5 — neural tube, gut and beating heart (even natural embryos stop at E11 in a dish).
- Dies without a placenta to connect to (engineered placentas for embryos do not exist yet).

# The end

Thank you for the attention!



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